



## 1. Description

### 1.1. Project

|                 |                   |
|-----------------|-------------------|
| Project Name    | EX7               |
| Board Name      | custom            |
| Generated with: | STM32CubeMX 6.3.0 |
| Date            | 09/15/2026        |

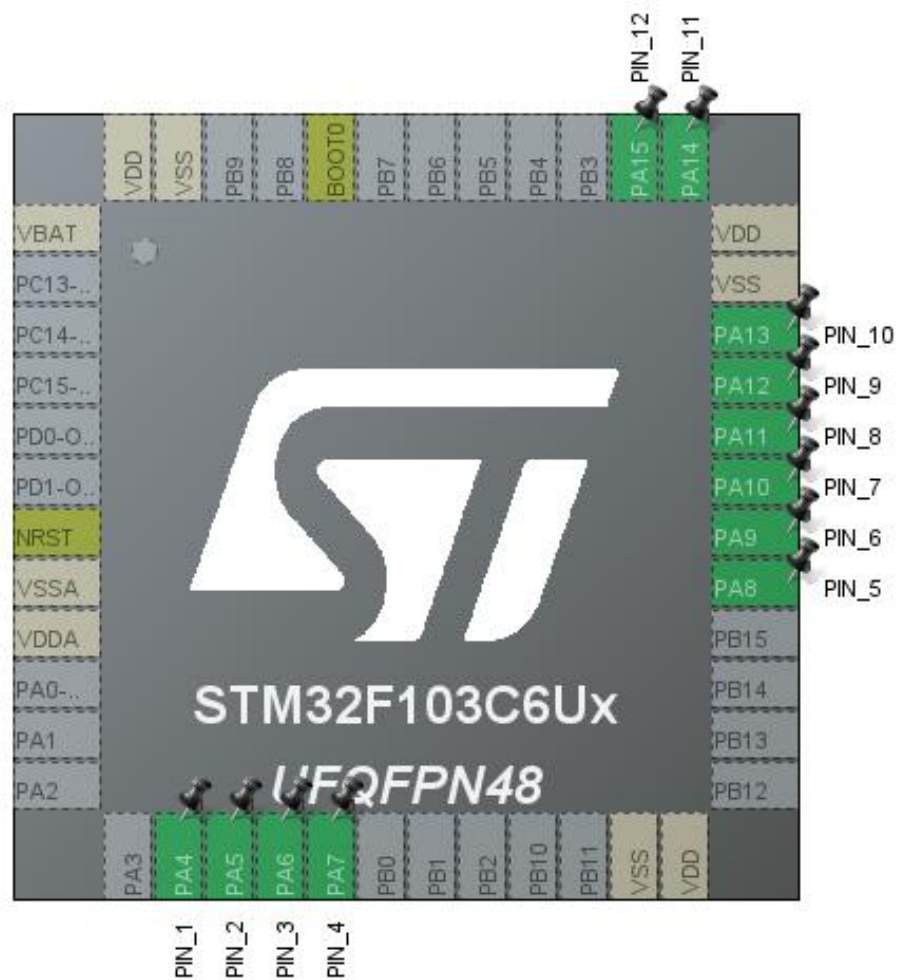
### 1.2. MCU

|                |               |
|----------------|---------------|
| MCU Series     | STM32F1       |
| MCU Line       | STM32F103     |
| MCU name       | STM32F103C6Ux |
| MCU Package    | UFQFPN48      |
| MCU Pin number | 48            |

### 1.3. Core(s) information

|         |               |
|---------|---------------|
| Core(s) | Arm Cortex-M3 |
|---------|---------------|

## 2. Pinout Configuration

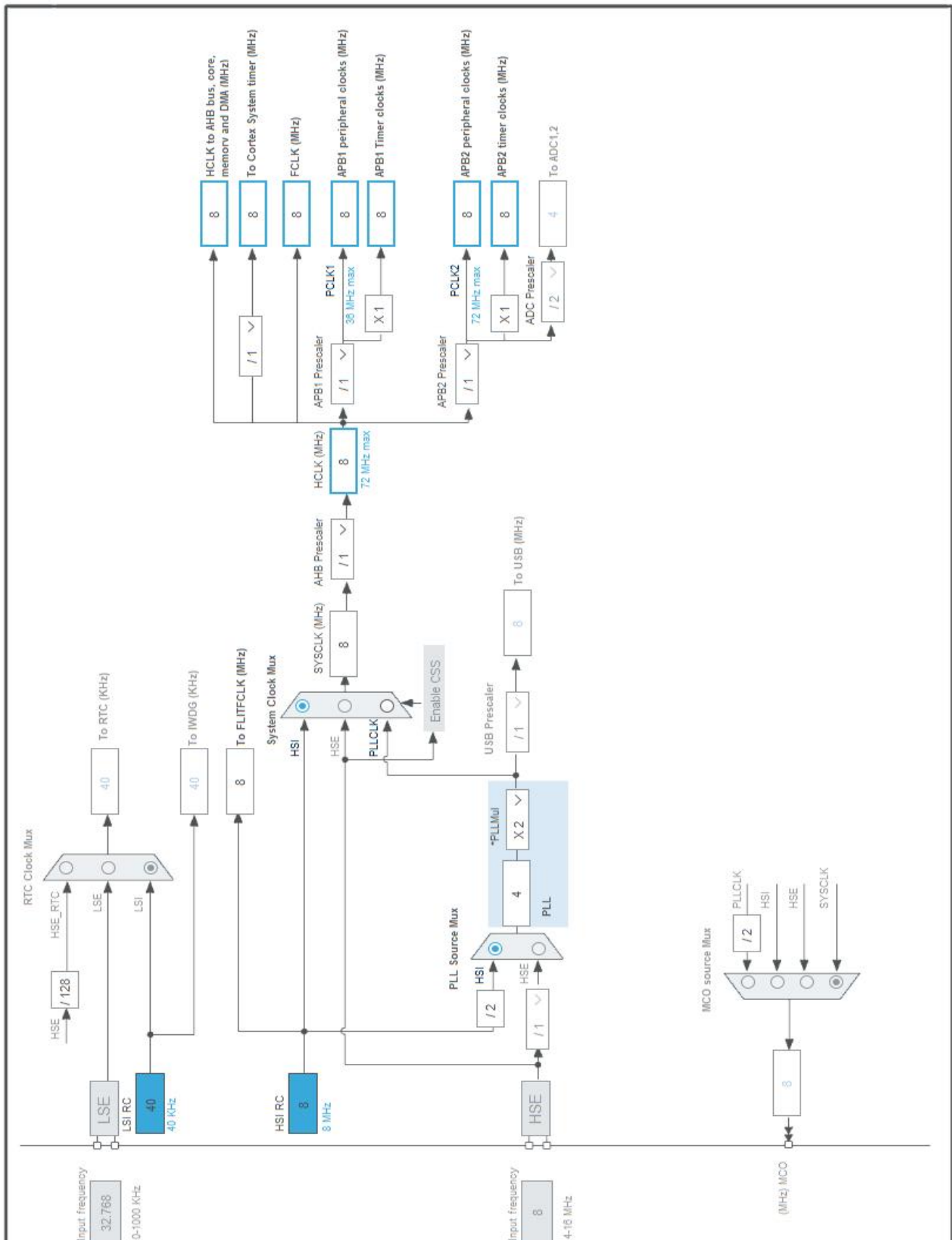


### 3. Pins Configuration

| Pin Number<br>UFQFPN48 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label  |
|------------------------|---------------------------------------|----------|--------------------------|--------|
| 1                      | VBAT                                  | Power    |                          |        |
| 7                      | NRST                                  | Reset    |                          |        |
| 8                      | VSSA                                  | Power    |                          |        |
| 9                      | VDDA                                  | Power    |                          |        |
| 14                     | PA4 *                                 | I/O      | GPIO_Output              | PIN_1  |
| 15                     | PA5 *                                 | I/O      | GPIO_Output              | PIN_2  |
| 16                     | PA6 *                                 | I/O      | GPIO_Output              | PIN_3  |
| 17                     | PA7 *                                 | I/O      | GPIO_Output              | PIN_4  |
| 23                     | VSS                                   | Power    |                          |        |
| 24                     | VDD                                   | Power    |                          |        |
| 29                     | PA8 *                                 | I/O      | GPIO_Output              | PIN_5  |
| 30                     | PA9 *                                 | I/O      | GPIO_Output              | PIN_6  |
| 31                     | PA10 *                                | I/O      | GPIO_Output              | PIN_7  |
| 32                     | PA11 *                                | I/O      | GPIO_Output              | PIN_8  |
| 33                     | PA12 *                                | I/O      | GPIO_Output              | PIN_9  |
| 34                     | PA13 *                                | I/O      | GPIO_Output              | PIN_10 |
| 35                     | VSS                                   | Power    |                          |        |
| 36                     | VDD                                   | Power    |                          |        |
| 37                     | PA14 *                                | I/O      | GPIO_Output              | PIN_11 |
| 38                     | PA15 *                                | I/O      | GPIO_Output              | PIN_12 |
| 44                     | BOOT0                                 | Boot     |                          |        |
| 47                     | VSS                                   | Power    |                          |        |
| 48                     | VDD                                   | Power    |                          |        |

\* The pin is affected with an I/O function

## 4. Clock Tree Configuration



## 5. Software Project

### 5.1. Project Settings

| Name                              | Value                                |
|-----------------------------------|--------------------------------------|
| Project Name                      | EX7                                  |
| Project Folder                    | F:\LabVXL\VXL_VDK_Lab\LAB1\EX7\STM32 |
| Toolchain / IDE                   | STM32CubeIDE                         |
| Firmware Package Name and Version | STM32Cube FW_F1 V1.8.7               |
| Application Structure             | Advanced                             |
| Generate Under Root               | Yes                                  |
| Do not generate the main()        | No                                   |
| Minimum Heap Size                 | 0x200                                |
| Minimum Stack Size                | 0x400                                |

### 5.2. Code Generation Settings

| Name                                                            | Value                                 |
|-----------------------------------------------------------------|---------------------------------------|
| STM32Cube MCU packages and embedded software                    | Copy only the necessary library files |
| Generate peripheral initialization as a pair of '.c/.h' files   | No                                    |
| Backup previously generated files when re-generating            | No                                    |
| Keep User Code when re-generating                               | Yes                                   |
| Delete previously generated files when not re-generated         | Yes                                   |
| Set all free pins as analog (to optimize the power consumption) | No                                    |
| Enable Full Assert                                              | No                                    |

### 5.3. Advanced Settings - Generated Function Calls

| Rank | Function Name      | Peripheral Instance Name |
|------|--------------------|--------------------------|
| 1    | SystemClock_Config | RCC                      |
| 2    | MX_GPIO_Init       | GPIO                     |

## 6. Power Consumption Calculator report

### 6.1. Microcontroller Selection

|           |               |
|-----------|---------------|
| Series    | STM32F1       |
| Line      | STM32F103     |
| MCU       | STM32F103C6Ux |
| Datasheet | DS5936 Rev7   |

### 6.2. Parameter Selection

|             |     |
|-------------|-----|
| Temperature | 25  |
| Vdd         | 3.3 |

### 6.3. Battery Selection

|                   |                 |
|-------------------|-----------------|
| Battery           | Li-SOCL2(A3400) |
| Capacity          | 3400.0 mAh      |
| Self Discharge    | 0.08 %/month    |
| Nominal Voltage   | 3.6 V           |
| Max Cont Current  | 100.0 mA        |
| Max Pulse Current | 200.0 mA        |
| Cells in series   | 1               |
| Cells in parallel | 1               |

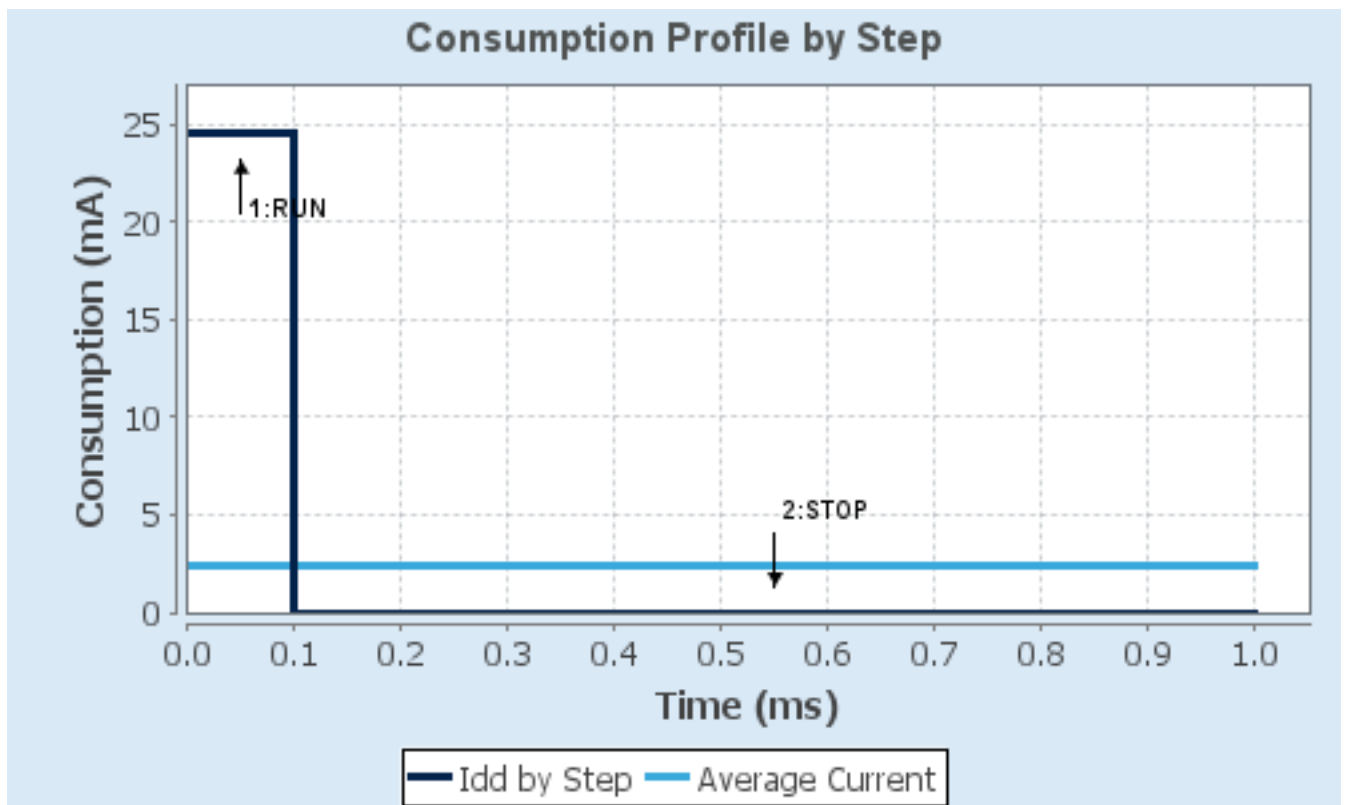
#### 6.4. Sequence

|                               |             |              |
|-------------------------------|-------------|--------------|
| <b>Step</b>                   | Step1       | Step2        |
| <b>Mode</b>                   | RUN         | STOP         |
| <b>Vdd</b>                    | 3.3         | 3.3          |
| <b>Voltage Source</b>         | Battery     | Battery      |
| <b>Range</b>                  | No Scale    | No Scale     |
| <b>Fetch Type</b>             | FLASH       | n/a          |
| <b>CPU Frequency</b>          | 72 MHz      | 0 Hz         |
| <b>Clock Configuration</b>    | HSE PLL     | Regulator_LP |
| <b>Clock Source Frequency</b> | 8 MHz       | 0 Hz         |
| <b>Peripherals</b>            |             |              |
| <b>Additional Cons.</b>       | 0 mA        | 0 mA         |
| <b>Average Current</b>        | 24.5 mA     | 11.7 $\mu$ A |
| <b>Duration</b>               | 0.1 ms      | 0.9 ms       |
| <b>DMIPS</b>                  | 90.0        | 0.0          |
| <b>Ta Max</b>                 | 102.41      | 105          |
| <b>Category</b>               | In DS Table | In DS Table  |

#### 6.5. Results

|               |                             |                 |            |
|---------------|-----------------------------|-----------------|------------|
| Sequence Time | 1 ms                        | Average Current | 2.46 mA    |
| Battery Life  | 1 month, 27 days,<br>1 hour | Average DMIPS   | 61.0 DMIPS |

#### 6.6. Chart



## 7. Peripherals and Middlewares Configuration

### 7.1. RCC

#### 7.1.1. Parameter Settings:

##### System Parameters:

|                   |                    |
|-------------------|--------------------|
| VDD voltage (V)   | 3.3                |
| Prefetch Buffer   | Enabled            |
| Flash Latency(WS) | 0 WS (1 CPU cycle) |

##### RCC Parameters:

|                                |      |
|--------------------------------|------|
| HSI Calibration Value          | 16   |
| HSE Startup Timeout Value (ms) | 100  |
| LSE Startup Timeout Value (ms) | 5000 |

### 7.2. SYS

**Debug: No Debug**

**Timebase Source: SysTick**

**\* User modified value**

## 8. System Configuration

### 8.1. GPIO configuration

| IP   | Pin  | Signal      | GPIO mode        | GPIO pull/up pull down      | Max Speed | User Label |
|------|------|-------------|------------------|-----------------------------|-----------|------------|
| GPIO | PA4  | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | PIN_1      |
|      | PA5  | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | PIN_2      |
|      | PA6  | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | PIN_3      |
|      | PA7  | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | PIN_4      |
|      | PA8  | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | PIN_5      |
|      | PA9  | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | PIN_6      |
|      | PA10 | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | PIN_7      |
|      | PA11 | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | PIN_8      |
|      | PA12 | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | PIN_9      |
|      | PA13 | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | PIN_10     |
|      | PA14 | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | PIN_11     |
|      | PA15 | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | PIN_12     |

### 8.2. DMA configuration

nothing configured in DMA service

### 8.3. NVIC configuration

#### 8.3.1. NVIC

| Interrupt Table                         | Enable | Preenmption Priority | SubPriority |
|-----------------------------------------|--------|----------------------|-------------|
| Non maskable interrupt                  | true   | 0                    | 0           |
| Hard fault interrupt                    | true   | 0                    | 0           |
| Memory management fault                 | true   | 0                    | 0           |
| Prefetch fault, memory access fault     | true   | 0                    | 0           |
| Undefined instruction or illegal state  | true   | 0                    | 0           |
| System service call via SWI instruction | true   | 0                    | 0           |
| Debug monitor                           | true   | 0                    | 0           |
| Pendable request for system service     | true   | 0                    | 0           |
| System tick timer                       | true   | 15                   | 0           |
| PVD interrupt through EXTI line 16      | unused |                      |             |
| Flash global interrupt                  | unused |                      |             |
| RCC global interrupt                    | unused |                      |             |

#### 8.3.2. NVIC Code generation

| Enabled interrupt Table                 | Select for init sequence ordering | Generate IRQ handler | Call HAL handler |
|-----------------------------------------|-----------------------------------|----------------------|------------------|
| Non maskable interrupt                  | false                             | true                 | false            |
| Hard fault interrupt                    | false                             | true                 | false            |
| Memory management fault                 | false                             | true                 | false            |
| Prefetch fault, memory access fault     | false                             | true                 | false            |
| Undefined instruction or illegal state  | false                             | true                 | false            |
| System service call via SWI instruction | false                             | true                 | false            |
| Debug monitor                           | false                             | true                 | false            |
| Pendable request for system service     | false                             | true                 | false            |
| System tick timer                       | false                             | true                 | true             |

\* User modified value

## 9. System Views

### 9.1. Category view

#### 9.1.1. Current

|            |  |  |  |  |
|------------|--|--|--|--|
| Middleware |  |  |  |  |
|------------|--|--|--|--|

| System Core                                                                              | Analog | Timers | Connectivity | Computing |
|------------------------------------------------------------------------------------------|--------|--------|--------------|-----------|
| DMA                                                                                      |        |        |              |           |
| GPIO  |        |        |              |           |
| I2C   |        |        |              |           |
| RCC   |        |        |              |           |
| SYS   |        |        |              |           |

## 10. Docs & Resources

| Type               | Link                                                                                                                                          |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Datasheet          | <a href="http://www.st.com/resource/en/datasheet/CD00210843.pdf">http://www.st.com/resource/en/datasheet/CD00210843.pdf</a>                   |
| Reference manual   | <a href="http://www.st.com/resource/en/reference_manual/CD00171190.pdf">http://www.st.com/resource/en/reference_manual/CD00171190.pdf</a>     |
| Programming manual | <a href="http://www.st.com/resource/en/programming_manual/CD00228163.pdf">http://www.st.com/resource/en/programming_manual/CD00228163.pdf</a> |
| Programming manual | <a href="http://www.st.com/resource/en/programming_manual/CD00283419.pdf">http://www.st.com/resource/en/programming_manual/CD00283419.pdf</a> |
| Errata sheet       | <a href="http://www.st.com/resource/en/errata_sheet/CD00211391.pdf">http://www.st.com/resource/en/errata_sheet/CD00211391.pdf</a>             |
| Application note   | <a href="http://www.st.com/resource/en/application_note/CD00160362.pdf">http://www.st.com/resource/en/application_note/CD00160362.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/CD00164185.pdf">http://www.st.com/resource/en/application_note/CD00164185.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/CD00167326.pdf">http://www.st.com/resource/en/application_note/CD00167326.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/CD00167594.pdf">http://www.st.com/resource/en/application_note/CD00167594.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/CD00211314.pdf">http://www.st.com/resource/en/application_note/CD00211314.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/CD00249778.pdf">http://www.st.com/resource/en/application_note/CD00249778.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/CD00259245.pdf">http://www.st.com/resource/en/application_note/CD00259245.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/CD00264321.pdf">http://www.st.com/resource/en/application_note/CD00264321.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/CD00264342.pdf">http://www.st.com/resource/en/application_note/CD00264342.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/CD00264379.pdf">http://www.st.com/resource/en/application_note/CD00264379.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/DM00024853.pdf">http://www.st.com/resource/en/application_note/DM00024853.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/DM00032987.pdf">http://www.st.com/resource/en/application_note/DM00032987.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/DM00033267.pdf">http://www.st.com/resource/en/application_note/DM00033267.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/DM00033344.pdf">http://www.st.com/resource/en/application_note/DM00033344.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/DM00042534.pdf">http://www.st.com/resource/en/application_note/DM00042534.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/DM00052530.pdf">http://www.st.com/resource/en/application_note/DM00052530.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/DM00073742.pdf">http://www.st.com/resource/en/application_note/DM00073742.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/DM00080497.pdf">http://www.st.com/resource/en/application_note/DM00080497.pdf</a>     |
| Application note   | <a href="http://www.st.com/resource/en/application_note/DM00129215.pdf">http://www.st.com/resource/en/application_note/DM00129215.pdf</a>     |

Application note [http://www.st.com/resource/en/application\\_note/DM00156964.pdf](http://www.st.com/resource/en/application_note/DM00156964.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00160482.pdf](http://www.st.com/resource/en/application_note/DM00160482.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00209695.pdf](http://www.st.com/resource/en/application_note/DM00209695.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00220769.pdf](http://www.st.com/resource/en/application_note/DM00220769.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00236305.pdf](http://www.st.com/resource/en/application_note/DM00236305.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00257177.pdf](http://www.st.com/resource/en/application_note/DM00257177.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00272912.pdf](http://www.st.com/resource/en/application_note/DM00272912.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00296349.pdf](http://www.st.com/resource/en/application_note/DM00296349.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00315319.pdf](http://www.st.com/resource/en/application_note/DM00315319.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00325582.pdf](http://www.st.com/resource/en/application_note/DM00325582.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00327191.pdf](http://www.st.com/resource/en/application_note/DM00327191.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00354244.pdf](http://www.st.com/resource/en/application_note/DM00354244.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00380469.pdf](http://www.st.com/resource/en/application_note/DM00380469.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00395696.pdf](http://www.st.com/resource/en/application_note/DM00395696.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00493651.pdf](http://www.st.com/resource/en/application_note/DM00493651.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00536349.pdf](http://www.st.com/resource/en/application_note/DM00536349.pdf)  
Application note [http://www.st.com/resource/en/application\\_note/DM00725181.pdf](http://www.st.com/resource/en/application_note/DM00725181.pdf)